MD. AKKHOR HASAN

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RESEARCH INTERESTS

Physics-informed machine learning for energy and process systems; battery degradation-mode diagnostics, identifiability and lifetime prognostics; neural-network surrogates for model predictive control; robustness and reproducibility of learned models (multi-seed and Monte Carlo evaluation); digital twins with hardware-in-the-loop deployment; autonomous robot navigation and control.

EDUCATION

B.Sc. in Mechatronics Engineering, Rajshahi University of Engineering & Technology (RUET), Bangladesh 2022 – 2026

- CGPA: **3.50/4.00**. Medium of instruction: English.
- Undergraduate Thesis: *"AI-Enhanced Digital Twin for Predictive Analytics of Battery Health and Degradation Dynamics"*. Supervisor: Sumaya Ishrat Moyeen, Assistant Professor, Dept. of Mechatronics Engineering, RUET. Accepted at IEEE CSDE 2026 (see Publication 4).
- Relevant coursework: Control Systems · Machine Intelligence · Robotics · Signal Processing · Automation (PLC, SCADA, PID) · Microprocessors · Data Analysis & Simulation · Fluid Mechanics · Manufacturing Processes.

Higher Secondary Certificate (HSC), Science, Govt. Edward College, Pabna 2020

- GPA: **5.00/5.00**.

Secondary School Certificate (SSC), Science, Pabna Zilla School, Pabna 2018

- GPA: **5.00/5.00**.

RESEARCH EXPERIENCE

Undergraduate Researcher, Dept. of Mechatronics Engineering, RUET 2024 – Present

Self-directed, sole-author research on physics-informed learning for battery health and process control, from problem formulation to manuscript: two preprints under review at Elsevier journals (Journal of Energy Storage, Journal of Process Control) and one in preparation. Thesis research carried out with co-authors under the supervision of Sumaya Ishrat Moyeen.

- **Battery degradation modes:** developed a physics-informed estimator that recovers degradation modes from partial discharge windows rather than full cycles; analysed which modes are identifiable from partial data; trained the model on simulation only and validated it on two cell chemistries (under review, *Journal of Energy Storage*; Publication 2).
- **Life forecasting and check-up design:** testing whether degradation-mode estimates improve battery life forecasts and the design of check-up tests compared with capacity alone, across three real-cell datasets (manuscript in preparation; Publication 3).
- **Learned surrogates for control:** built physics-informed and black-box neural-network surrogates of a nonisothermal CSTR and placed them inside a model predictive control loop; evaluated nominal tracking across 7 random seeds and robustness through Monte Carlo analysis (under review, *Journal of Process Control*; Publication 1).
- **Battery digital twin (thesis):** trained and compared seven SOH models (including GRU–XGBoost, LSTM and CNN variants) under leave-one-battery-out cross-validation on NASA lithium-ion aging data, then deployed the best, XGBoost, hardware-in-the-loop in MATLAB/Simulink with automatic DC cooling triggered above a surface-temperature threshold (Paper from this thesis accepted at IEEE CSDE 2026, first author with S. R. Oishy and S. I. Moyeen; Publication 4).

PUBLICATIONS

Journal preprints (under review)

1. **M. A. Hasan**, "Physics-informed neural network surrogates for model predictive control of a nonisothermal CSTR: A multi-seed evaluation of nominal tracking and robustness," submitted to *Journal of Process Control*, Elsevier. Preprint available on SSRN. **[Sole author] [Preprints]**
2. **M. A. Hasan**, "Physics-informed estimation of battery degradation modes from partial discharge windows: Identifiability, simulation-only training and validation on two chemistries," submitted to *Journal of Energy Storage*, Elsevier. Preprint available on SSRN. **[Sole author] [Preprints]**

Manuscripts in preparation

3. **M. A. Hasan**, "Do degradation-mode estimates improve battery life forecasts and check-up design? A test against capacity alone on three real-cell datasets," in preparation for *Journal of Energy Storage*. **[Sole author]**

Conference papers (accepted, not yet published)

4. **M. A. Hasan**, S. R. Oishy, and S. I. Moyeen, "AI-enhanced digital twin with hardware-in-the-loop deployment for real-time lithium-ion battery state of health estimation," accepted at *IEEE CSDE 2026*, Paper ID: ieee-csde_4711. Presented in 1st November 2026. **[First author]**

SELECTED PROJECTS

Autonomous Quadrotor Navigation and SLAM Sep 2026

- MATLAB simulation of a quadrotor with full 6-DOF dynamics, LiDAR-based mapping and localisation, and navigation under stochastic wind disturbances. [Github]

Mobile Robot Navigation Stack **Sep 2026**

- Global path planning with Theta*, path tracking with Pure Pursuit, and local obstacle avoidance with artificial potential fields; completed test runs with zero collisions. [Github]

Real-Time Tracking and Stabilization System **2023**

- Python/OpenCV object detection and tracking coupled to a tuned PID controller; evaluated rise time, overshoot and steady-state error under dynamic conditions.

Multi-DOF Industrial Robotic Arm Prototype **2023**

- Designed the mechanical assembly in SolidWorks and programmed inverse kinematics on Arduino for pick-and-place tasks.

Smart DC Motor with Autonomous Speed Control **2022**

- Closed-loop PWM speed control holding a target RPM under varying mechanical load; documented response characteristics.

TECHNICAL SKILLS

Programming: Python, MATLAB/Simulink, C, C++.
Machine learning: TensorFlow, Keras, PyTorch (familiar), scikit-learn, XGBoost; physics-informed neural networks, recurrent networks (GRU), time-series feature engineering, uncertainty and robustness evaluation.
Control & modelling: model predictive control, PID tuning, state-space modelling, battery electrochemical/degradation modelling, hardware-in-the-loop testing.
Robotics & vision: OpenCV, path planning (Theta*, Pure Pursuit, APF), SLAM concepts, Arduino, ESP32.
Automation & CAD: PLC (Ladder, FBD, SFC, ST), SCADA, HMI, SolidWorks, AutoCAD.
Tools & languages: Git, LaTeX, MS Office; English (professional; all coursework, thesis and manuscripts written in English), Bengali (native).

AWARDS AND HONOURS

- **Best Officer:** TEDxRUET 2023–24 Committee.
- **Semifinalist:** Hult Prize Regional Business Case Competition, RUET.
- **Runner-Up:** Idea Competition, Tech Mayhem, Robotic Society of RUET.
- **Semifinalist:** ROBO Wrestling, ROBOTRONICS 2.0, Robotic Society of RUET.

WORKSHOPS AND TRAINING

Fundamentals of Internet of Things, **RUET IoT Club** · 3D Printing & Introduction to Design Engineering, **Society of Computer Aided Designers, RUET (2022)** · UI & UX Design, **Code Studio.**

INDUSTRIAL AND PROFESSIONAL EXPERIENCE

Industrial Trainee, GPH Ispat Ltd., Bangladesh **2023**

- Hands-on training in automated production lines, PLC and electrical control systems, substation operations, and preventive and predictive maintenance at a large steel plant.

Video Editor, AI Content & Workflow Operations Specialist, AutomateX, Rajshahi **Sep 2024 – Aug 2026**

- Built and quality-checked generative-AI production workflows for US-based clients; tracked performance metrics and kept structured English documentation across time zones.

LEADERSHIP AND SERVICE

Chief Technical Officer, Radio RUET (2022–2025) · Organizing Secretary, MTE Career Club, RUET (2023–2024) · Video & Production Lead, TEDxRUET (2024–2025) · Director of Cinematography, Hult Prize at RUET (2024–2025) · Vice President, Ovijatrik-RUET Adventure Club (2023–2025) · Vice President, Pabna District Association, RUET · Campus Ambassador, EXPEDITIOUS 2023, Robo Mechatronics Association Bangladesh.

REFERENCES

Md. Firoz Ali, Head, Dept. of Mechatronics Engineering, RUET ;
Sumaya Ishrat Moyeen, Assistant Professor and Thesis Supervisor, Dept. of Mechatronics Engineering, RUET . Further contact details available upon request.